

AI BLOG FACTORY

Digital guides for students, creators & tech learners

THE AI CODING COMPANION

Learn Programming Faster with AI as Your Personal Tutor

By Joy Roy

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A Note from the Author

I wrote this book as a patient mentor. Every technique here was tested in real conditions — as a student, with real deadlines. Nothing here is theoretical.

Let us get to work.

-- Joy Roy

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Why I Wrote This Book

Personal story from Joy. When I started learning Python, I was stuck on errors for hours. Then I learned to use AI as a tutor. This changed everything.

The journey from frustration to building with AI — and how you can start today.

1. The Old Way

Stare at error. Google. Stack Overflow. Try random fixes. Hours of frustration. Move on.

- Start with this technique in your current course
- Practice for 15 minutes daily throughout the week
- Note what works and what does not

QUICK WIN

Try this today: Take one why i wrote this book prompt from this chapter and use it on a real task.

2. The New Way

Paste the error to AI. Get a clear explanation. Understand the cause. Fix it. Move on in minutes.

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- Practice for 15 minutes daily throughout the week
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ACTION

Before moving to the next chapter, implement one why i wrote this book technique in your current work.

Setting Up

One afternoon. Three tools. A full AI-assisted development environment.

VS Code. GitHub Copilot. ChatGPT. All free. All ready.

1. VS Code

The editor. Free. Add the Python extension.

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- Practice for 15 minutes daily throughout the week
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QUICK WIN

Try this today: Take one setting up prompt from this chapter and use it on a real task.

2. GitHub Copilot

Free for students via education.github.com. AI that writes code as you type.

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- Practice for 15 minutes daily throughout the week
- Note what works and what does not

ACTION

Before moving to the next chapter, implement one setting up technique in your current work.

3. ChatGPT or Claude

Your tutor. Available 24. Patient. Never judges bad questions.

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The S-O-L-V-E Framework

A five-step process for every coding problem. Use this and you will never be stuck for more than 10 minutes.

The complete debugging loop: State → Observe → Locate → Verify → Execute.

1. S — State

What should the code do? What does it actually do?

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QUICK WIN

Try this today: Take one the s-o-l-v-e framework prompt from this chapter and use it on a real task.

2. O — Observe

Copy the FULL error message. Every line matters.

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ACTION

Before moving to the next chapter, implement one the s-o-l-v-e framework technique in your current work.

3. L — Locate

Which line caused it? What is the error type?

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4. V — Verify

Paste everything. Ask for a simple explanation.

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5. E — Execute

Apply the fix. Test it. If it fails, the AI response will tell you why.

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Your First 8 Projects

Build as you learn. AI helps when you are stuck. The goal is to ship, not to be perfect.

Eight progressive projects from beginner to portfolio-ready.

1. Project 1: Calculator

Variables, operators, input/output.

- Start with this technique in your current course
- Practice for 15 minutes daily throughout the week
- Note what works and what does not

QUICK WIN

Try this today: Take one your first 8 projects prompt from this chapter and use it on a real task.

2. Project 2: To-Do List

Lists, loops, functions.

- Start with this technique in your current course
- Practice for 15 minutes daily throughout the week
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ACTION

Before moving to the next chapter, implement one your first 8 projects technique in your current work.

3. Project 3: CSV Analyzer

Pandas, data cleaning, basic visualization.

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4. Project 4: Data Dashboard

Streamlit, matplotlib, real data.

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5. Project 5: Web Scraper

Requests, BeautifulSoup, data extraction.

- Start with this technique in your current course
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- Note what works and what does not

6. Project 6: Quiz App

Logic, conditionals, scoring.

- Start with this technique in your current course
- Practice for 15 minutes daily throughout the week
- Note what works and what does not

7. Project 7: API Integration

JSON, APIs, real-world data.

- Start with this technique in your current course
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8. Project 8: Prediction Model

scikit-learn basics, a real ML pipeline.

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Portfolio Power

Three projects on GitHub. That is how you get interviews.

Clean READMEs. Deployed demos. The portfolio that gets callbacks.

1. README Template

Title, description, how to run, what it does, what you learned.

- Start with this technique in your current course
- Practice for 15 minutes daily throughout the week
- Note what works and what does not

QUICK WIN

Try this today: Take one portfolio power prompt from this chapter and use it on a real task.

2. Clean Code

Comments. Meaningful variable names. Small functions. AI helps review your code.

- Start with this technique in your current course
- Practice for 15 minutes daily throughout the week
- Note what works and what does not

ACTION

Before moving to the next chapter, implement one portfolio power technique in your current work.

3. Deployment

Streamlit Community Cloud for free. Deploy your dashboard. Share the link.

- Start with this technique in your current course
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Quick-Reference Cheat Sheet

Print this. Stick it next to your workspace.

- 1. Why I Wrote This Book
- 2. Setting Up
- 3. The S-O-L-V-E Framework
- 4. Your First 8 Projects
- 5. Portfolio Power

About the Author

Joy Roy is the founder of AI Blog Factory — a studio creating premium digital guides for students, creators, and tech learners. A BSc Data Science student who built everything in this book while studying, not after graduating.

AI Blog Factory — Digital guides for students, creators & tech learners.

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